

ChronoMind: A Neural Framework for Logical and Temporal Consistency Verification in Short Stories

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Abstract—ChronoMind’s automated framework analyzes short fiction narratives for logical and temporal accuracy. To assess coherence, each section of a narrative is evaluated based on its logical consistency and temporal ordering. To perform this task, machine learning is used to classify narrative coherence, producing outputs from a transformer-based model along with contextual interpretation of the narrative. The system is developed using a multi-year dataset (2016–2018) consisting of approximately 101,307 narrative segments, and utilizes a fine-tuned RoBERTa transformer model under supervised learning to determine whether sections of narratives are coherent or incoherent.

Experimental results validate the effectiveness of the proposed approach, achieving 97.41% accuracy, 0.9745 F1-score, and 0.994 ROC-AUC, indicating strong generalization and classification performance. In addition to coherence classification, the system incorporates a contextual tone analysis module that identifies stylistic characteristics such as neutrality, irony, and sarcasm. This module provides complementary information that helps identify subtle inconsistencies in narrative flow. The use of transformer-based contextual representations enables automated verification of logical and temporal consistency across narrative segments, allowing the system to capture document-level dependencies and detect inconsistencies that extend beyond individual sentences.

Index Terms—Narrative Coherence Classification, Temporal Consistency Verification, Logical Consistency Detection, Transformer-Based Models, RoBERTa, Literary Tone Analysis, Contextual Language Modeling, Short Story Analysis, Neural Text Evaluation, Natural Language Processing (NLP).

I. INTRODUCTION

When stories are told digitally, they can have varying degrees of quality and/or consistency, which is becoming increasingly important for automated systems to measure. Both the logical sequencing of events and the timing of events should be considered by all writers during the story development phase. Inconsistencies in sequencing of events, development of characters, and contextual understanding of the story will create potential barriers to the reader’s comprehension of the story. Because narrative relationships exist at a level that is greater than an individual sentence and often rely on outside contextual information that may not be immediately apparent to the reader, it has been difficult to automatically identify narrative inconsistencies within a short story in an efficient or effective manner. Most NLP technologies focus only on

sentence-level interaction and/or rules that govern authoring using sentence-level interactions and they do not function well when attempting to understand the structure of the underlying story due to their inability to comprehend multiple interrelationships between different sentences. Research involving transformer-based language models for narrative frameworks has increased dramatically over the past few years, however, more applied studies are needed in order to generate an effective method for developing more standardized forms of assessment for narrative structures.

To solve these issues, this work proposes ChronoMind; a neural framework for verifying coherence between the events occurring within a short narrative series by applying template-based reasoning. The proposed method includes a method of fine-tuning a RoBERTa-based transformer model for coherence classification through supervised learning on a structured multi-year data set. Additionally, a tone analysis software module has been added to allow for a contextual reading of narrative qualities of the text such as neutrality, irony, and sarcasm in order to aid both in interpreting the coherence of the narrative and also to understand how to best classify it. Experimental results indicate that transformer architectures have excellent generalizability in the context of narrative evaluation and for intelligent content-analysis applications.

II. LITERATURE SURVEY

Li et al. created the first large, human-annotated dataset called CONTRADOC, which they developed to enable the detection of internal contradictions within large documents using large language models (LLMs). They used document-based methods to identify contradictions, whereas other current methods use natural language inference (NLI) to analyze two sentences. CONTRADOC contains three types of texts: news articles, stories, and Wikipedia articles. Additionally, it classifies multiple types of self-contradiction by employing code types of Numerically Inconsistent (NI), Factually inconsistent (FI), Logically inconsistent (LI), Emotionally inconsistent (EI), and Perspective Inconsistent (PI). The authors tested how well LLMs (e.g., GPT-3.5, GPT-4, PaLM-2, LLaMA-2) were able to identify contradictions and employed two different evaluation approaches (i.e., binary and evidence-based) to measure the self-contradictions. The new generation of

advanced LLMs cannot find contradictions sufficiently as they lack contextual knowledge to understand self-contradicting evidence (Tracking the contradictions requires tracking many sentences or pages worth of information). Through the trial, GPT-4 is superior at locating self-contradictions when compared to other LLMs.

The introduction of new features in the development of Transformer architecture has enhanced natural language processing methods. This is due to the ability of the Transformer to enable machine processing of language over very long periods of time. The publication titled “Attention is All You Need” provided an alternative to both Recurrent and Convolutional models via the introduction of a family of self-attention mechanisms that can be applied in parallel across a full sequence and maintain a global relationship of the tokens; rather than only maintaining a local relationship to adjacent tokens. The end result of this new architecture was to improve the scalability of the computation and mitigate many of the issues related to sequential processing. Analyzing narratives involves researchers establishing logical and temporal relationships, which depend on how the various elements of a story interact with each other over the course of the entire narrative. Traditional recurrent models do not adequately capture what is called a “distributed” or “long-range” dependency such that the various elements of a narrative can interact across large distances. The dynamic nature of the Transformer’s layers provides a mechanism for the model to assign attention weights to tokens, allowing for a more complete understanding of all paragraphs of text. This shift in architecture has laid the groundwork necessary to produce today’s modern text processing models that are the basis for the systems used to perform coherence classification and narrative verification.

BERT enhanced transformer architecture by employing deep bidirectional contextual representations through the use of masked language modeling. BERT uses a bidirectional model to encode information from both adjacent and following token positions at the same time. The use of a bidirectional model allows BERT to encode more context, therefore providing a better understanding of semantics. The use of a bidirectional model improved performance on many NLP tasks, such as text classification, inferences, and language understanding. Narrative verification is heavily reliant on creating coherence between sentences that are not adjacent; thus, the use of BERT’s bidirectional model is critical for creating coherence in larger text samples. BERT exhibited exceptional generalization performance; however, BERT’s original next-sentence prediction task showed deficiencies concerning the modeling of a document’s overall continuity. BERT defined transfer learning for all subsequent NLP tasks that employed transformer architecture, thus serving as a foundation for the subsequent improvements made by later transformer architectures.

There have been many changes made with respect to BERT’s original pre-training approach, including alterations

to how next-sentence prediction was performed as well as implementing dynamic masking and using larger datasets/batch sizes during training. The importance of choosing an appropriate optimization technique was highlighted by these changes because they demonstrate the major role played by the choice of optimizer in determining how well transformers perform compared to other types of models that have successfully achieved strong performance across multiple evaluation tasks related to natural language processing (NLP). The high level of performance achieved by this type of pre-training scheme can largely be explained by the high generalization capabilities of neural networks when fine-tuning the package with datasets containing large amounts of contextual information necessary for the correct interpretation of narrative structure. Finally, the RoBERTa framework can serve as an effective approach for developing neural network models which verify linguistic and temporal consistency across multiple, short stories.

The aim of the manner is to use timeliness to make language models more capable of reasoning about temporal relationships by providing a structure for consistency. In their paper “Learning to Reason Over Time: Timeline Self-Reflection for Temporal Reasoning with Language Models,” the authors suggest that language models should validate their inferences by verifying event sequences against (intermediate) timelines. Furthermore, these researchers found that large language models produced temporally ambiguous data as they attempted to infer different events occurring at various points along the same narrative timeline. They designed their model in such a way as to represent timeliness as an intermediate representation so they could assess how accurately they predicted events. They showed statistically significant improvement in temporal consistency on their ability to verify events on numerous benchmark datasets. The authors also pointed out that context-based embeddings provide insufficient temporal consistency across multiple timelines based on different temporal references and do not offer any architecture specifically aimed at addressing this issue. Because of the length and importance of the short narratives, they require a substantial degree of timeliness to evidence the sequencing of events from one time to another.

III. METHODOLOGY

A. Problem Formulation

ChronoMind is a tool developed to test a narrative’s logical coherence and temporal consistency by using two techniques: neural classification of coherence; and an analysis of the contextual tones of the narrative’s segments. The main goal of this tool is to determine if a narrative segment contains semantic consistency and structured continuity throughout the different sentences in the segment. In short narrative stories, inconsistencies can occur because of an abrupt change in events, a contradiction of descriptive language, or a break in the flow of the narrative. Therefore, the problem is addressed as a supervised classification, whereby the model learns representations of the context of the segments to capture their

interrelatedness, event relationships, and implicit temporal cues among all the sentences in the narrative. In contrast to sentence level classification, the need for this type of modeling arises from the need to model structural consistency across the entire narrative segment. Formally, let x denote a narrative text segment and y represent its coherence label. The task is defined as a binary classification problem:

$$y \in \{0, 1\} \quad (1)$$

where $y = 1$ indicates a coherent narrative segment and $y = 0$ indicates an incoherent segment. The model learns a function

$$f(x) \rightarrow y \quad (2)$$

that maps contextual embeddings of the input text to its corresponding consistency label.

B. Dataset and PreProcessing

The ChronoMind model is evaluated based on a rich collection of narrative texts that have been gathered through the span of three years, between 2016 and 2018. These narratives involve around 101,307 sequences of narrative texts drawn from short stories, where each sequence represents a sequence of events that needs to be coherent according to the logical and chronological order. Coherence can be defined as the ordering of events that conforms with the rest of the story world. Incoherence results when there exist contradictions between events in a particular narrative. In order to address label bias in the evaluation process carried out by humans, more than one evaluation was performed, with label validation taking place via reaching a consensus among evaluators.

During the preprocessing step, raw narrative text is prepared for subsequent learning via transformers in an optimal format. In this regard, the first stage involves text cleaning, where extraneous noise and tokens that do not serve the purpose are stripped off from the raw text to ensure standardization throughout. Next, RoBERTa tokenizer applies byte-level subword tokenization in order to handle the context properly and also address any out-of-vocabulary words in the text. Special tokens needed by the transformer architecture, including classification tokens, separator tokens, are automatically added to each sequence. Since input sequences need to be uniform in terms of lengths to enable batch processing, all sequences are standardized to a predetermined length by applying padding and truncation techniques. Furthermore, attention masks are created for each input to help differentiate between actual tokens and padded tokens. In addition, labels are created for each narrative text based on whether or not the narrative is incoherent, resulting in binary targets (0 for incoherent narratives and 1 for coherent narratives).

C. Neural Architecture

To classify narrative coherence, ChronoMind leverages a transformer-based neural network structure that has been developed based on the RoBERTa-base architecture. The primary reason for using RoBERTa as opposed to the original BERT is its improved pre-training methods compared to BERT through

the use of dynamic masking, removal of next sentence prediction and the use of much larger and more diverse datasets during training. As a result, RoBERTa is much more capable of representing and understanding contextual relationships in a text. When performing analyses of narratives, coherence is made possible by the relationships between sentences or events which requires accurate long distance representations of the underlying data. As such, RoBERTa's architecture provides excellent contextual representation and generalization across various narrative types.

The byte-level BPE tokenizer in RoBERTa operates at the subword level without losing any original context of the words. The subword tokens are processed through a series of multi-layer transformer networks w/ multi-head self-attention and feedforward networks. Multi-head self-attention allows flexibility within the model to assign varying levels of importance to the tokens relative to one another. Capturing many different types of semantic, logical and temporal relationships based upon the surrounding context of events within a token. Unlike traditional sequential based models, the transformer processes all tokens in parallel; therefore, the transformer is able to effectively model local interactions and global dependencies through the model. Upon full completion of processing the entire segment of a narrative, the models create an embedding that contains the full structural and meaning of a narrative.

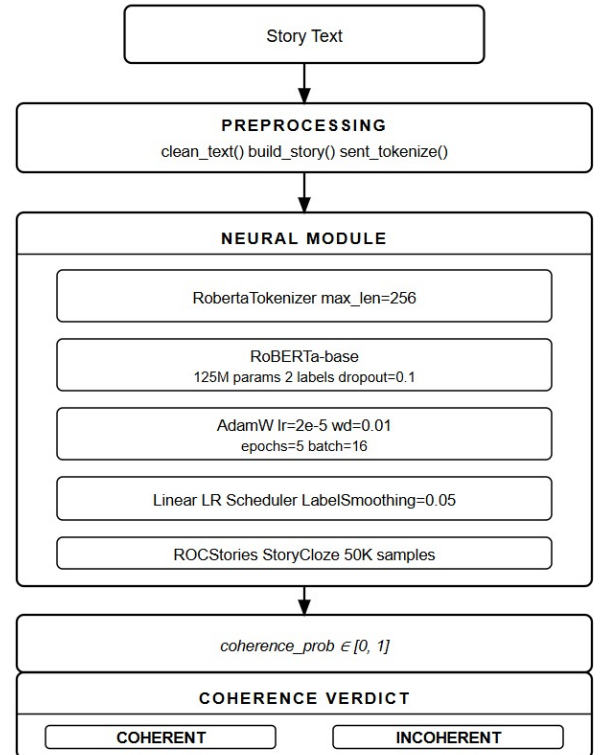


Fig. 1. ChronoMind architecture.

After extracting the contextual representations associated with the narrative portion, a set of classifications is processed

through a classification layer. A fully connected classification layer will convert the high-dimensional representations into a fixed-length representation that can be used to generate predictions. The fixed-length representation is then mapped to a probability distribution, i.e., to a probability distribution across the two coherence classes: coherent and incoherent. Thus, using this mapping, the model can identify inconsistencies in event ordering, logical continuation, and narrative progression in the document as a whole. Therefore, coherence assessments for both sentences and whole documents can be made by using the contextual representation.

$$\hat{y} = \text{Softmax}(Wh + b) \quad (3)$$

In this schematic, h is the contextual embedding vector for the input narrative segment, W is the weight matrix from the classification layer, and b is the bias vector from the classification layer. The vector output $\hat{y}$ represents a probability distribution over coherence class; therefore, it indicates how likely a narrative segment is to be coherent and incoherent. The model is optimized using the cross-entropy loss function:

$$\mathcal{L} = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (4)$$

The model learns using a cross-entropy cost function (which measures the difference between predicted and actual labels) to guide the learning process. Optimal convergence of the model is achieved with AdamW Optimizer 2×10^{-5} to maintain stable convergence and avoid overfitting. The proposed training configuration will enable the model to learn significant patterns from narrative data while still providing adequate generalization capabilities/test performance.

D. Training Strategy and Optimization

The training process of ChronoMind involves the utilization of RoBERTa architecture framework to develop a coherence classification model using the supervised learning method. During the training process, we carried out optimizations and validations to ensure that we attain a generalized and converged model.

The training process has been conducted for five epochs. However, validation checkpoints are introduced throughout the training process with the intention of ensuring that our model has been trained and avoid the risk of overfitting. Validation checkpoints will be important to select our best model based on validation accuracy since there is a higher likelihood that our best model will be generalized when predicting unseen data.

We opted to use the AdamW optimization technique because of the advantages it gave us when applying decoupled weight decay, which improves our model’s generalization. We adopted a learning rate scheduler with the aim of making the optimization process stable by introducing a warm-up phase in the initial stages of training, which we anticipated would make gradual changes to our model’s parameters. It means that we

anticipated that any form of fluctuation caused by instability would be minimized.

Label smoothing has been done during the training process. This technique will be important since it will improve the accuracy of the model’s predictions by having some probability mass allocated to several classes. The learning rate is defined as:

$$\eta = 2 \times 10^{-5} \quad (5)$$

Label smoothing modifies the target distribution as:

$$y^{LS} = (1 - \epsilon)y + \frac{\epsilon}{K} \quad (6)$$

where $\epsilon = 0.05$ is the smoothing factor and K is the number of classes. Mini-batch gradient descent is used for optimization, and the objective function minimized during training is:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta) \quad (7)$$

where θ represents the model parameters. The best-performing model is selected based on validation accuracy and saved for final evaluation. GPU acceleration is used to improve computational efficiency, and training and validation metrics are recorded at each epoch to analyze convergence behavior.

E. Tone Analysis Module

In an attempt to make its assessment go beyond mere structural consistency, ChronoMind incorporates a tone analysis component into the evaluation process to better understand context through narratives. Coherence classification focuses on identifying logical consistency in events while tone analysis provides additional data relating to the style and expression of the texts. Tone plays a critical role in how a narrative is perceived because tone helps us understand the events and characters portrayed. In some cases, inconsistent tones such as the sudden change from straightforward to ironic or sarcastic narrative tones indicate that something in the plot is out of place. Such changes in narrative tone may arise from character inconsistency, emotional inconsistency, or narrative inconsistencies which are not always identified via structural or event-based coherence assessments. Hence, incorporating tone analysis allows the program to be able to identify implicit contextual clues as well as stylistic variations in a story.

This module is designed using a transformer-based approach, wherein contextualized embeddings are used for the narrative segments. The network computes contextualized embeddings for each of the input segments by capturing not only the semantics but also the style associated with the text. The embeddings generated are further passed through a classification layer that predicts the tone of the text segment into one of the predefined classes such as neutral, ironic, or sarcastic. The tone prediction function is defined as:

$$\hat{t} = \text{Softmax}(W_t h + b_t) \quad (8)$$

In such a setup, the W_t and b_t indicate the weight and bias matrix corresponding to the tone classification layer, respectively, and the $\hat{t}$ indicates the probability distribution across the defined tone classes. This arrangement allows for the learning

of more complex style-related characteristics and nuances of language usage, beyond the simple sentiment polarity aspect. Unlike existing sentiment analysis techniques, where the issue of sentiment polarity is usually taken care of, in the current technique, there are more fine-grained narrative features which could affect perceptions regarding coherence. Tone classification combined with the process of coherence verification by ChronoMind ensures convenience and robustness of tasks.

F. Evaluation Metrics

There are several evaluation standards used to measure ChronoMind's overall classification performance depending on specific classifications included in its output. One example would be the coherence verification task as a binary classification task in which we can examine ChronoMind's classification performance by looking at 5 specific metrics: accuracy, precision, recall, F1 Score, and ROC-AUC, each of which provides insight into different aspects of ChronoMind's overall classification performance (Accuracy), the ability to accurately classify examples by distinguishing the two classes (Precision, Recall, and F1), and the robustness of the classification error to classifying examples (ROC-AUC). 1. Accuracy: The system will classify a segment as coherent based on whether it was accurately classified by cross referencing the test segment against its corresponding reference segments. 2. Precision: This value is the total number of segments classified as coherent, but incorrectly labeled as such in relation to the total number of segments, which have been assigned a coherent label. 3. Recall: This value will identify the number of segments assigned as coherent, based on whether a segment is coherent or not. 4. F1 Score: F1 average accuracy will be employed to calculate the overall basic accuracy of test segments divided by both false positive and false negative results from misclassifying segments during evaluation. 5. ROC-AUC: This measure relies upon the model's effectiveness in differentiating between consistent and inconsistent outputs, across all possible classifications of true positive classifications to true negative classifications within the system. The evaluation metrics are formally defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives respectively.

G. Results and Discussion

As per the confusion matrix, both classifications of narratives have only a few misclassifications. In addition, there is a good balance of performance between the coherent and incoherent narrative classifications, demonstrating the ability of the model to generalize to unseen data. The AUC value of the ROC curve approaches an AUC of 1.0, indicating that there is excellent discrimination of the classes at different decision thresholds within the upper left corner of the ROC curve.

Moreover, both the training and validation accuracies show stability in learning with little overfitting. Thus, the proposed transformer-based model has successfully captured temporal and contextual relationships in the narratives. The transformer approach has provided greater accuracy compared to traditional sentence-based approaches for classifying narrative coherence.

H. Algorithm description

The ChronoMind system follows a structured pipeline to evaluate narrative coherence and tone. The algorithmic workflow is described as follows:

- Step 1: Input a narrative text segment x .
- Step 2: Perform preprocessing using text cleaning, sentence tokenization, and normalization.
- Step 3: Convert the processed text into token embeddings using the RoBERTa tokenizer.
- Step 4: Pass the tokenized input into the RoBERTa transformer model to obtain contextual embeddings.
- Step 5: Apply a classification layer to compute the coherence probability score y using softmax.
- Step 6: Perform tone analysis using a classification model to compute tone probabilities t .
- Step 7: Evaluate the model using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
- Step 8: Output the final coherence classification and tone prediction.

IV. CONCLUSION

The proposed framework operates with the use of transformer technology which evaluates the logical and temporal coherence of short stories by applying the modified version of the RoBERTa model capable of using contextual information in different parts of the narrative to ensure an accurate coherence assessment. The system detects chronological event, logical, and narrative progression errors since it works with deep contextual embeddings. In such a way, the suggested framework will serve as a reliable approach to evaluating the consistency of narrations on the basis of the presented evidence.

The analysis framework introduces tonal analysis, which allows analyzing not only individual structural components of a narration but also all minor stylistic devices narrators use to show their intention via neutral, sarcastic, or ironic tones. The proposed system gains the capability of recognizing narrative inconsistency since it analyzes whole narrations rather than individual sentences, which increases the likelihood of getting

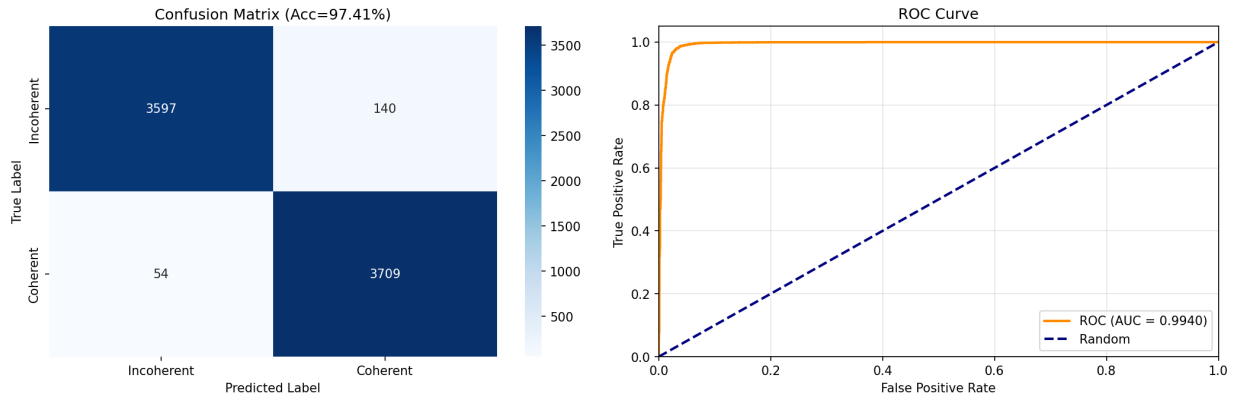


Fig. 2. ChronoMind: Neural Evaluation. Confusion matrix and ROC curve showing coherence classification performance with 97.41% accuracy and 0.994 ROC-AUC.

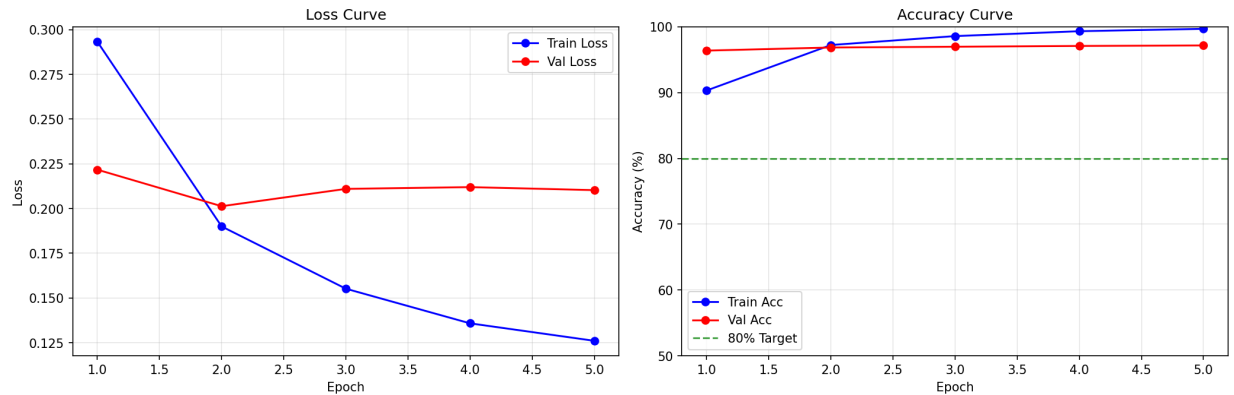


Fig. 3. ChronoMind: Training Curves showing loss and accuracy progression across epochs.

accurate results. Moreover, it can be said that the suggested system has great potential because ChronoMind demonstrated outstanding results with its accuracy of 97.41

Therefore, it should be said that the objectives of creating consistent narrations through classification and temporal consistency detection were achieved due to the application of transformer architectures that assess narrations in terms of their coherence and chronological sequence.

REFERENCES

- [1] “CONTRADOC: Understanding Self-Contradictions in Documents with Large Language Models,” Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), 2024.
- [2] Attention Is All You Need, A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” Advances in Neural Information Processing Systems (NeurIPS), 2017.
- [3] BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” Proceedings of NAACL-HLT, 2019.

- [4] RoBERTa: A Robustly Optimized BERT Pretraining Approach, Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “RoBERTa: A robustly optimized BERT pretraining approach,” arXiv preprint arXiv:1907.11692, 2019.

- [5] Narrative-of-Thought: Improving Temporal Reasoning of Large Language Models via Recounted Narratives, Authors, “Narrative-of-Thought: Improving temporal reasoning of large language models via recounted narratives,” Findings of EMNLP, 2024.

- [6] Learning to Reason Over Time: Timeline Self-Reflection for Improved Temporal Reasoning in Language Models, Authors, “Learning to Reason Over Time: Timeline self-reflection for improved temporal reasoning in language models,” arXiv preprint, 2025.

- [7] NeSTR: A Neuro-Symbolic Abductive Framework for Temporal Reasoning in Large Language Models, Authors, “NeSTR: A neuro-symbolic abductive framework for temporal reasoning in large language models,” arXiv preprint, 2025.

- [8] SCORE: Story Coherence and Retrieval Enhancement for AI Narratives, Authors, “SCORE: Story coherence and retrieval enhancement for AI narratives,” arXiv preprint, 2025.

- [9] Dont Stop Pretraining: Adapt Language Models to

Domains and Tasks, S. Gururangan, A. Marasović, S. Swayamdipta, K. Lo, I. Beltagy, D. Downey, and N. A. Smith, “Don’t stop pretraining: Adapt language models to domains and tasks,” Proceedings of ACL, 2020.

[10] SemEval Irony and Sarcasm Detection Tasks, Task Organizers, “SemEval shared tasks on irony and sarcasm detection,” SemEval, 2018.